Smart Agro System for Crop Recommendation Using Deep Learning

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***Abstract*—Agriculture is significantly affected by climate and soil characteristics such that traditional crop selection methods do not take into account the changes occurring in the environment on a daily basis. With the development of machine learning and deep learning techniques, intelligent systems can be created to analyze agricultural data sourced from multiple locations, enabling well-informed decision-making for crop selection. This paper introduces an integrated smart weather-agro system that incorporates convolutional neural networks to classify soils based on images captured using cameras. In addition, the system provides recommendations for the most suitable crops based on soil nutrient levels and current live weather conditions. A method has been developed to classify soils into distinct categories using imagery data and to recommend crops based on nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall. Live weather data is continuously obtained through a weather API to update the recommendations, ensuring their relevance over time. The results demonstrate a strong ability to accurately classify various soil types using images and generate highly reliable crop recommendations aligned with agronomic patterns. A web-based application developed using Flask makes the system practical for real-life agricultural scenarios. Overall, this study highlights the effectiveness of combining deep learning, environmental information, and web technologies for precision agriculture.**

***Index Terms*—smart agriculture, soil classification, convolutional neural networks, crop recommendation, machine learning, weather-aware systems**

I. Introduction

Farmers are crucial for global food security, and the success of agriculture also impacts a country's economy, especially in developing nations [1]. A large portion of the economy of most developing countries is derived from crop production; therefore, crop yield is important to these economies [5]. However, farmers face many challenges when making agricultural decisions due to uncertainty with many different factors such as weather conditions, soil properties, and inefficient use of limited resources [4]. For many years, farmers relied only on their own experience to guide their decision-making, but due to the rapidly changing conditions within the environment, farmers can no longer depend solely on this historical knowledge base.

The emergence of intelligent agriculture technologies has led to the development of advanced agricultural systems that employ machine learning and deep learning techniques to analyze large volumes of complex agricultural data and extract meaningful patterns [1], [6]. Many existing crop recommendation systems primarily rely on numerical representations of soil and climatic factors, such as precipitation, temperature, and soil nutrient levels, to determine suitable crops for cultivation [4], [5].

Despite the increasing use of machine learning in agriculture, many existing crop recommendation systems rely primarily on numerical soil parameters obtained from laboratory testing or structured datasets. Such systems often cannot utilize visual soil characteristics that may provide additional insights into soil composition and structure [7]. Furthermore, several approaches treat soil analysis, weather monitoring, and crop recommendation as independent tasks rather than components of a unified intelligent agricultural framework. To address these limitations, this work proposes an integrated smart agro-system that combines image-based soil classification using deep learning with ensemble-based crop recommendation and real-time weather data integration. By unifying these components within a single decision-support platform, the proposed system aims to provide more practical and context-aware agricultural recommendations.

The major contributions of this work are summarized as follows:

- Image-based soil classification using a Convolutional Neural Network (CNN) to automatically identify soil types from captured soil images.
- Integration of soil characteristics and environmental parameters such as nitrogen (N), phosphorus (P), potassium

(K), temperature, humidity, pH, and rainfall to generate crop recommendations.

- Comparative benchmarking of multiple machine learning models to identify the most effective algorithm for the crop recommendation task.
- Development of a web-based decision-support system that allows users to upload soil images, view weather conditions, and obtain suitable crop recommendations.
- Support for manual input of soil nutrient parameters (N, P, K, and pH) when laboratory soil testing results are available, improving recommendation accuracy.

While our previous work focused primarily on CNN-based soil classification for identifying soil types from images, the present study extends that work by integrating soil image analysis with a complete smart agriculture decision-support system. The proposed system combines soil classification, crop recommendation based on agronomic parameters (N, P, K, pH, temperature, humidity, and rainfall), real-time weather data integration, and a web-based user interface for practical deployment. This integrated framework enables farmers to obtain crop recommendations by combining visual soil characteristics with environmental conditions, thereby providing a more comprehensive agricultural advisory system compared to earlier CNN-based soil classification work.

II. System Architecture

The Smart Weather Agro System is designed using a modular microservices-based architecture that supports real-time prediction and scalability [6], [8]. The system consists of several interconnected components, including data acquisition and preprocessing, machine learning models, and visual presentation layers.

The architecture accepts inputs from users in the form of soil images and geographic location data [7]. A trained convolutional neural network classifier is used to classify soil types based on the provided images. Simultaneously, real-time weather parameters such as temperature, humidity, and rainfall are collected through a weather application programming interface [12]. These weather parameters are integrated with soil nutritional content to determine suitable crops for cultivation [4], [5].

The modular framework of the Smart Weather Agro System enables separation of data processing, model execution, and user interaction components. This design enhances system efficiency, reliability, security, scalability, and extensibility, making it suitable for real-world agricultural deployment.

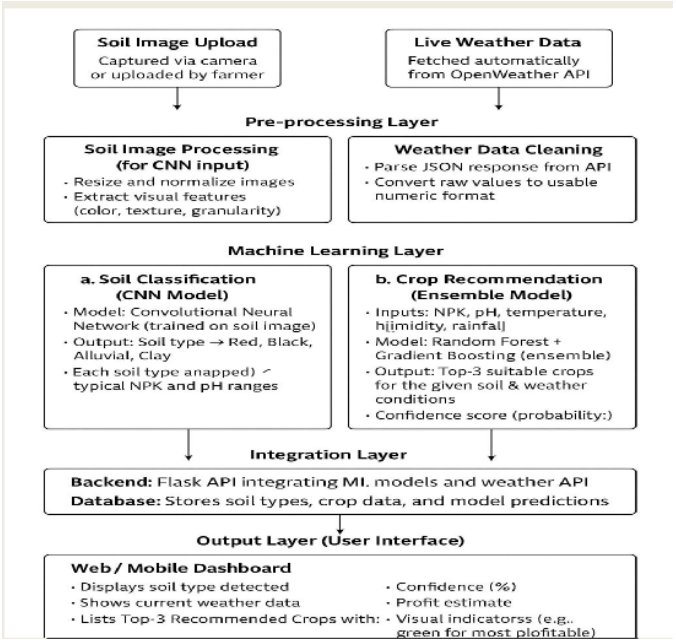


Fig. 1. System architecture

III. Data Sources and Preprocessing

A. Soil Image Dataset

The soil image dataset used in this study was compiled from publicly available sources. Approximately 90% of the images were obtained from the open-source Soil Classification Dataset available on GitHub [8], while additional images were collected from publicly available web sources to improve dataset diversity. The final dataset contains images from five soil categories: alluvial (363 images), black (522 images), laterite (295 images), red (588 images), and yellow (769 images).

To improve dataset quality and diversity, duplicate and visually similar images were identified and removed during preprocessing using image hashing techniques. Similar images found within the dataset were replaced with newly collected soil images obtained from public web sources to increase variability and reduce redundancy. All images were resized to 128×128 pixels and normalized before being used for training the CNN model. This normalization ensures consistent input dimensions and enables the Convolutional Neural Network (CNN) to effectively learn spatial features required for accurate soil type classification.

The final dataset contains a total of 2537 soil images.

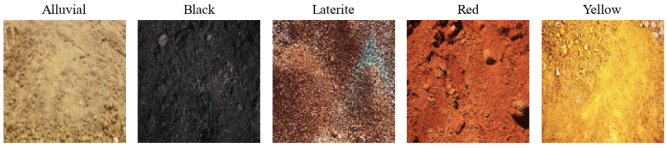


Fig. 2. Sample soil images

For model evaluation, the dataset was divided into training and validation subsets. The CNN model was trained using the training set, while validation data was used to monitor model performance and prevent overfitting during training. Additional unseen soil images collected from publicly available web sources were also used to assess the generalization ability of the trained model.

B. Crop Recommendation Dataset

The crop recommendation model was trained using the Crop Recommendation Dataset published by Atharva Ingle on Kaggle [4], which contains 2200 samples with agronomic features including nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall, along with 22 crop classes. These features represent soil nutrient levels and environmental conditions that influence crop suitability. Min-max scaling was applied to normalize the feature values before model training [7].

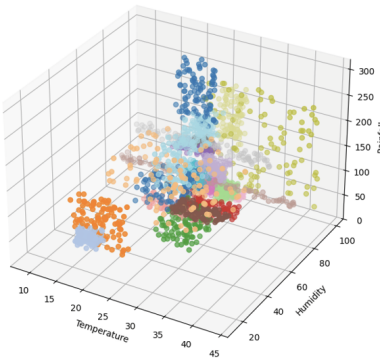


Fig. 3. 3D climate distribution of crops

C. Weather Data Acquisition

An external weather API is used to obtain real-time weather data for the selected city. Parameters such as temperature, humidity, and rainfall are dynamically retrieved from the API, allowing the system to generate crop recommendations based on current climatic conditions. In situations where the API service is unavailable, the system uses predefined fallback values to ensure continuous operation.

In future implementations, the system can be extended by integrating

IoT-based environmental sensors for field-level monitoring [10] and direct collection of weather parameters such as temperature, humidity, rainfall, and soil moisture. Such integration would enable more accurate and localized environmental monitoring, thereby improving the reliability of crop recommendations.

These environmental parameters are combined with soil nutrient information to evaluate crop suitability under current climatic conditions. By considering both soil and weather characteristics, the system provides more context- aware and reliable crop recommendations.

The classification of crops according to their respective temperatures and amounts of rainfall produces several distinct crop distribution clusters arising from similar crop types. For example, rice and maize are found primarily in areas of high rainfall, whereas pulses, including lentils and chickpeas, will typically occur in areas of moderate rainfall. These data demonstrate how closely related climatic variables impact a crop's ability to grow, and they provide support for employing weather variables within a recommended cropping system.

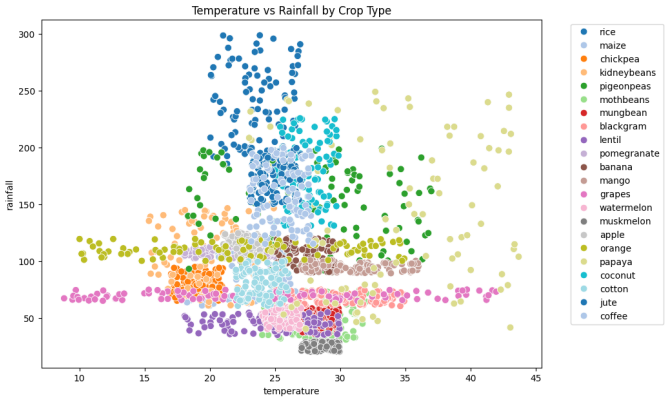


Fig. 4. Temperature vs Rainfall by crop type

IV. EXPLORATORY DATA ANALYSIS

A. Feature Correlation Analysis

Correlation analysis was performed to examine relationships among soil nutrients and environmental parameters. The correlation heatmap reveals a strong positive correlation between phosphorus (P) and potassium (K), with a coefficient of approximately 0.74, indicating potential multicollinearity. Such relationships are important to identify, as correlated features can influence machine learning model behavior.

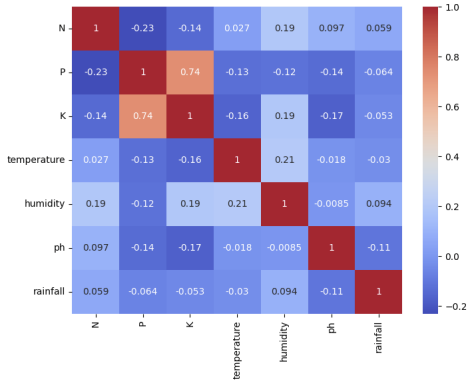


Fig. 5. Feature Correlation Heatmap

The other environmental and nutrient features tested (i.e., temperature, humidity, pH, and rainfall) showed that they have weaker correlations and thus can be considered independent contributors to crop growth.

B. Comparative Benchmarking of Machine Learning Models

As shown in Table 1, multiple machine learning models were evaluated for the crop recommendation task. Ensemble-based approaches such as Random Forest, Bagging, and Gradient Boosting demonstrated strong performance in the agricultural dataset.

Based on this comparative evaluation, Random Forest and Gradient Boosting were selected to form a voting-based ensemble model. Random Forest provides strong robustness and stability by aggregating multiple decision trees, while Gradient

TABLE I
COMPARATIVE BENCHMARKING OF MACHINE LEARNING MODELS FOR CROP RECOMMENDATION

Model	Accuracy (%)
Gaussian Naïve Bayes	**99.55**
Random Forest	**99.32**
Bagging Classifier	98.86
Decision Tree	98.18
Gradient Boosting	98.18
Support Vector Machine (SVM)	96.82
Logistic Regression	96.36
K-Nearest Neighbors (KNN)	95.91
Extra Tree Classifier	92.95
AdaBoost	14.55

Boosting improves predictive performance by sequentially correcting the errors of previous models. By combining the predictions of these two models using a soft voting mechanism,

$$P(y = c) = \frac{1}{M} \sum_{m=1}^{M} P_m(y = c) \tag{1}$$

Where M = number of classifiers

The ensemble approach improves generalization and provides more reliable crop recommendations for varying soil and environmental conditions.

C. Pairwise Feature Distribution Analysis

Pairwise distribution analysis indicates that crops cluster differently based on their temperature and rainfall range. While some crops prefer a narrow range of environmental conditions, others can tolerate a much broader range of conditions. The non-linear relationships between temperature and rainfall mean that using machine learning models, rather than a rules-based model, makes more sense.



Fig. 6. Pairwise feature distribution analysis of the crop dataset

V. CNN-BASED SOIL CLASSIFICATION

A. Model Architecture

Soil classification is performed using a Convolutional Neural Network (CNN), which is well suited for image recognition tasks due to its ability to automatically learn spatial features from image data [2], [3]. The proposed CNN architecture consists of convolutional layers, pooling layers, and fully connected layers to classify soil images into different soil categories.

The important parameters and configurations of the CNN model used for soil classification include:

- Input image size: $128 \times 128 \times 3$
- Two convolutional layers with 32 and 64 filters using 3×3 kernels
- Max-pooling layers for spatial dimensionality reduction
- ReLU activation function for intermediate layers
- Dropout layers with rates of 0.3 and 0.5 to prevent overfitting
- Fully connected dense layer with 128 neurons
- Softmax activation function for the output layer
- L2 regularization with factor 1×10^{-4}

The model is trained using the Adam optimizer with categorical cross-entropy loss and accuracy as the evaluation metric. Early stopping and learning rate scheduling techniques are applied during training to improve generalization and stabilize the learning process.

B. Training and Validation Performance

The accuracy for both training and validation is continually improved through the training process, with training accuracy achieving about 97% and validation accuracy stabilizing between 93% - 95%. The loss curves show a consistent reduction toward zero, indicating a positive impact of regularisation and normalisation during the model's training process and indicating an acceptable level of overfitting.

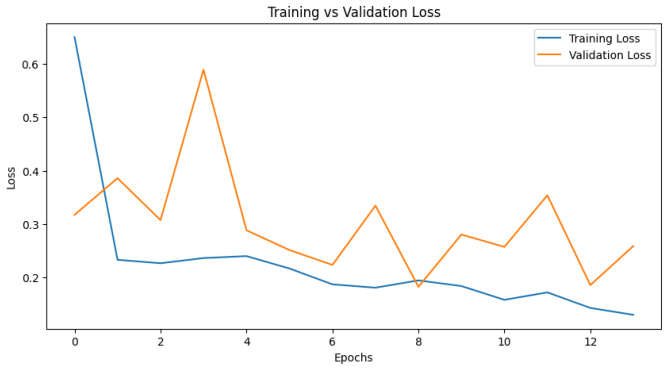


Fig. 7. Training vs Validation Loss

VI. CROP RECOMMENDATION METHODOLOGY

A. Ensemble Learning for Crop Recommendation

Machine learning techniques are widely used for crop recommendation by analyzing relationships between soil nutrients

and environmental conditions [5], [7]. In this study, ensemble-based learning methods are employed to capture the nonlinear relationships among important agronomic parameters such as nitrogen (N), phosphorus (P), potassium (K), soil pH, temperature, humidity, and rainfall.

Specifically, Random Forest and Gradient Boosting classifiers are utilized to model the complex interactions between soil nutrient levels and climatic conditions. Ensemble models are particularly suitable for agricultural prediction tasks because they combine multiple decision trees to improve prediction accuracy and reduce overfitting.

The outputs of the Random Forest and Gradient Boosting models are aggregated to compute a final crop suitability score based on the confidence of each model's prediction. This ensemble strategy improves the reliability of crop recommendations by reducing bias associated with individual models.

B. Soil Nutrient Profile Mapping

To incorporate soil characteristics into the crop recommendation model, representative nutrient profiles were assigned to each soil class identified by the CNN-based classifier. These profiles include nitrogen (N), phosphorus (P), potassium (K), and soil pH values, expressed in kg/ha for nutrient concentrations. The values represent typical agronomic ranges for common Indian soil types such as alluvial, black, laterite, red, and yellow soils. The estimates were derived from publicly available soil databases and agricultural studies, including SoilGrids, the Harmonized World Soil Database (HWSD), and ICAR agricultural reports [13]–[15]. These representative nutrient vectors are used to construct normalized feature vectors that are subsequently combined with real-time weather parameters for crop recommendation.

These representative values are used only as approximate soil profiles to support the crop recommendation process and are not intended to replace laboratory-based soil nutrient analysis.

C. Similarity-Based Recommendation

In addition to ensemble learning, a similarity-based recommendation mechanism is used to identify crops that closely match the given soil and environmental conditions. The feature vector includes soil nutrient values (N, P, K), soil pH, and weather parameters such as temperature, humidity, and rainfall [4], [5].

The input feature vector is first normalized using min-max scaling and then compared with historical crop data in the dataset. Similarity between the input vector and dataset samples is computed using the Euclidean distance measure [7].

$$d = \|X_{scaled} - x_{input}\|_2 \tag{2}$$

Crops corresponding to the smallest distance values are selected as the most suitable recommendations. The system presents the top three recommended crops to the user along with their confidence scores and estimated profitability.

VII. EVALUATION METRICS

A. Accuracy

Accuracy measures the overall correctness of the soil classification model by computing the ratio of correctly predicted soil images for the total number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

where

- TP represents true positives,
- TN represents true negatives,
- FP represents false positives,
- FN represents false negatives.

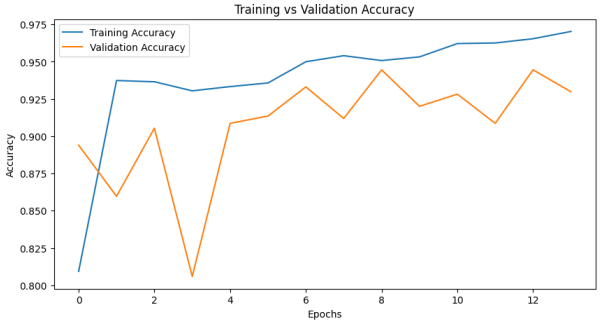


Fig. 8. Training vs Validation Accuracy

B. Confusion Matrix

The confusion matrix provides a detailed class-wise evaluation of prediction outcomes by comparing actual soil labels with predicted labels. For a multi-class soil classification problem, the confusion matrix is defined as:

$$C_{ij} = |\{x \mid y(x) = i \wedge \hat{y}(x) = j\}| \tag{4}$$

where

- $y(x)$ denotes the true class label,
- $\hat{y}(x)$ denotes the predicted class label,
- C_{ij} represents the number of samples belonging to a class i that are predicted as that class j.

There is a significant presence of strong diagonal dominance seen within the confusion matrix for soil sample classification, which indicates very few of the soil samples (i.e., over 90%) are correctly categorized according to their respective categories. The presence of very low off-diagonal values indicates that the soil types, such as alluvial, black, clay, and red soils, tend to be only minimally misclassified with other soil types. This supports the theory that CNN models are a solid foundation for training on soil samples [6], [11].

C. ROC Curve

An ROC curve assesses a classifier's discriminative power based on the relationship between true positive rates (TPR) and false positive rates (FPR). TPR and FPR are defined as

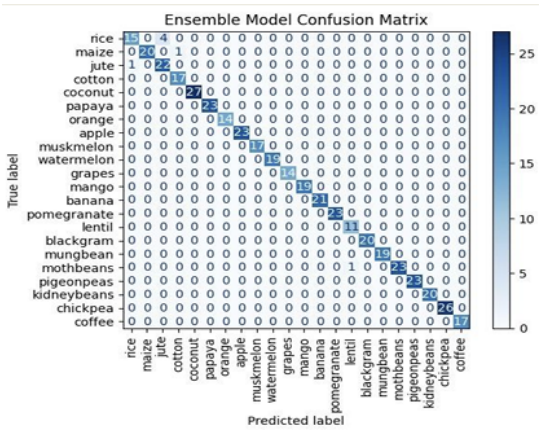


Fig. 9. Confusion Matrix of Soil Classification Results

$$\text{TPR} = \frac{TP}{TP + FN} \tag{5}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{6}$$

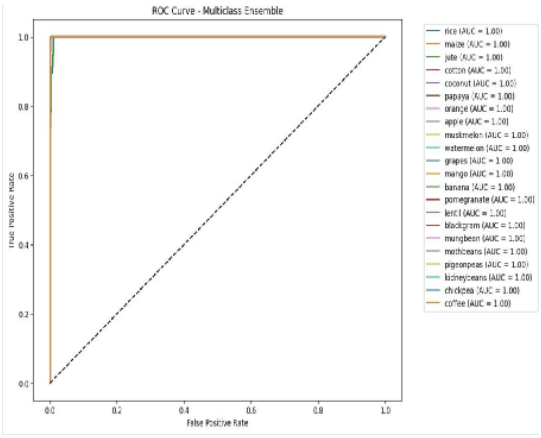


Fig. 10. ROC curve

The ROC curve is a visual representation of the TPR and FPR for various levels of thresholds for a classifier. The AUC is a measurement that gives a numerical value based on how well the classification model performs as measured by how high or how low it will value between 1 and 0 on the AUC scale. If the AUC value is high, such as if it approaches 1, then this indicates that the classifier has the capability of separating the various classes of data in its classification of all types of soil within the dataset.

VIII. Results and User Interface

Users can upload soil photos and choose their location using the developed system's interactive web-based user interface. The system shows the classified soil type, suggested crops, confidence scores, and pertinent meteorological parameters based on the input data. Easy interaction for practical agricultural use is made possible by the interface's intuitive and user-friendly design.

In addition to soil image-based analysis, the system also allows users to manually input soil nutrient parameters such as nitrogen (N), phosphorus (P), potassium (K), and pH values if reliable laboratory soil testing results are available. This option enables the system to generate more accurate crop recommendations based on reliable soil measurements.

While still supporting fast image-based estimation by analyzing soil images and mapping them to the most probable soil type and associated parameters for crop recommendation, the system particularly benefits farmers who lack access to laboratory soil testing facilities because of financial constraints or limited infrastructure availability.

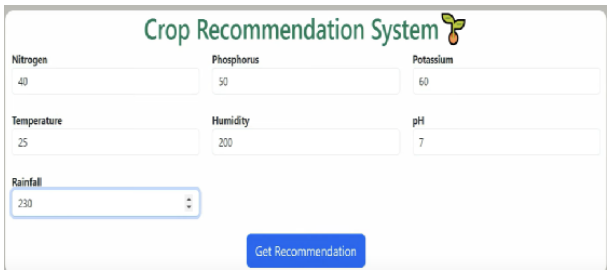


Fig. 11. Web-based user interface for crop recommendation

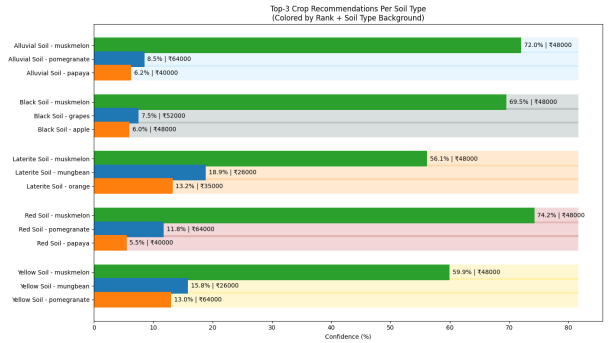


Fig. 12. Crop Recommendation Output

In connection with soil and present trends, the system will evaluate which three crops contain the best likelihood for being grown in the specific types of soils from which the user has selected and will provide a level of confidence and projected rate of return for each suggestion made. Supporting documentation indicates that muskmelon has the greatest level of confidence across all five soil types, as the normalized representation of the soil and the normalized weather trends most closely matched the requirements of the muskmelon crop when trying to grow the crop. Other recommended crops include pomegranate, grape, mung bean, and papaya; however, the levels of suitability for these crop types are based upon the given characteristics of each of the soils. The results illustrate the ability of this system to tailor the recommended crop types to the restrictions within the user's specific soil type.

In addition, through the inclusion of economic data in the selection of the recommended crops, the user's ability to make better agronomic management decisions is further enhanced by the information provided on the projected profitability of the various crop suggestions.

IX. IMPLEMENTATION AND DEPLOYMENT

A. Limitations

The system shows great potential for farming but also has certain limitations. These include the limited number of soil images available in the dataset, dependence on internet connectivity for real-time weather updates, and reliance on estimated soil nutrient profiles unless manual soil measurements are provided.

B. Future Scope

Future research will focus on integrating Internet-of-Things (IoT) soil-sensing technology to monitor nutrients in real time, further improving recommendation accuracy through advanced ensemble learning strategies, expanding region-specific datasets to increase model robustness, and enabling offline edge deployment.

Another potential direction is the integration of mobile applications and remote sensing data, such as satellite-based environmental observations, to provide large-scale monitoring and more precise crop recommendations for farmers.

X. CONCLUSION

This paper outlines the development of a Smart Weather-Agro System that utilizes both machine learning and deep learning technologies in conjunction with time-sensitive weather data for improving agricultural decision-making through intelligent automation and data-driven analysis. By using both image classification techniques and environment analysis tools for soil types, the proposed Smart Weather/Agro System is capable of providing farmers with accurate and actionable recommendations regarding their crops. The performance of the proposed system has been validated through experiments demonstrating the high potential of the proposed Smart Weather-Agro System as a valuable tool in digital agriculture.

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